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# Advancing Emotional Health Assessments: A Hybrid Deep Learning Approach Using Physiological Signals for Robust Emotion Recognition

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**ABSTRACT** Emotional health significantly impacts physical and psychological well-being, with emotional imbalances and cognitive disorders leading to various health issues. Timely diagnosis of mental illnesses is crucial for preventing severe disorders and enhancing medical care quality. Physiological signals, such as Electrocardiograms (ECG) and Electroencephalograms (EEG), which reflect cardiac and neuronal activities, are reliable for emotion recognition as they are less susceptible to manipulation than physical signals. Galvanic Skin Response (GSR) is also closely linked to emotional states. Researchers have developed various methods for classifying signals to detect emotions. However, these signals are susceptible to noise and are inherently non-stationary, meaning they change constantly over time. Consequently, emotions can vary rapidly. Traditional techniques for analyzing physiological signals may not be adequate to study the dynamic changes in emotional states. This research introduces a deep learning approach using a combination of advanced signal processing and machine learning to analyze physiological signals for emotion recognition. We propose a CNN-Vision Transformer (CVT) based method with ensemble classification. The process involves decomposing signals into segments, removing noise, and extracting features using 1D CNN and Vision Transformers. These features are integrated into a single vector for classification by an ensemble of LSTM, ELM, and SVM classifiers. The outputs are then synthesized using Model Agnostic Meta Learning (MAML) to improve prediction accuracy. Validated on AMIGOS and DEAP datasets with 10-fold cross-validation, our method achieved accuracies up to 98.2%, sensitivity of 99.15%, and specificity of 99.53%, outperforming existing emotion charting techniques. This novel method provides significant improvements 3 to 4% in the accuracy, sensitivity, and specificity of emotion detection, leveraging physiological signals for comprehensive emotional assessments.

**INDEX TERMS** Emotion Charting, Emotion Recognition, 1D CNN, Vision Transformer (ViT), Model Agnostic Meta Learning (MAML), Long-Short Time memory (LSTM), Extreme Machine Learning (ELM)

## I. INTRODUCTION

Affective computing via physiological signals is a rapidly evolving field having numerous applications; therefore, it has gained significant attention, particularly within the healthcare industry [1]. Emotion recognition is considered one of the finest domains of affective computing, and it has gotten great

attention in recent years. Emotions are the representation of human intuitions and human mental state [2]. Emotions are recognized using both human physiological and non-physiological signals. In recent research, there is an increasing interest in using physiological signals, such as heart rate, respiration, skin impedance, and Electroencephalogram

(EEG), for emotion recognition because physiological signals are more difficult to fake than human physical signals such as facial expressions. The physiological signals such as EEG, temperature (T), electrocardiogram (ECG), electromyogram (EMG), galvanic skin response (GSR), and respiration (RSP) are internal to the human body and are generated from the human central nervous system [1]. Figure 1, 2, and 3 shows some of the samples of EEG, ECG, and GSR from the AMIGOS dataset [3].

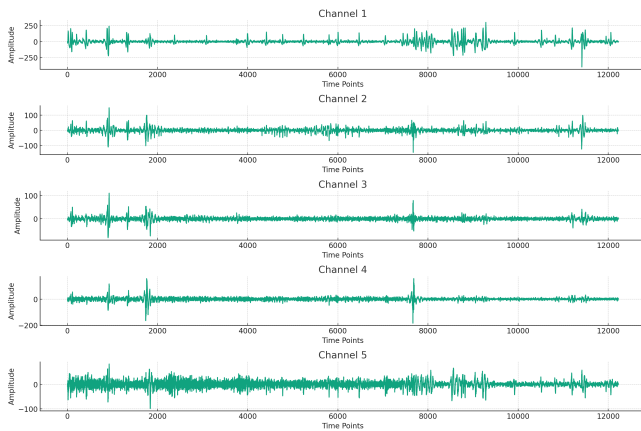


FIGURE 1: AMIGOS dataset samples of EEG signal

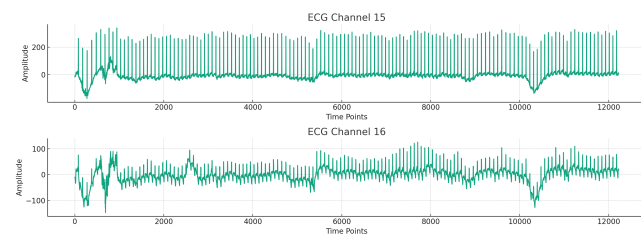


FIGURE 2: AMIGOS dataset samples of ECG signal

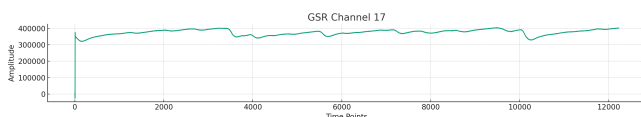


FIGURE 3: AMIGOS dataset samples of GSR signal

A typical method for emotion recognition using physiological signals consists of preprocessing, feature extraction and classification [4]. Preprocessing generally involves the removal of noise caused by cross-talk, body movement, and/or electromagnetic interference using different signal processing techniques. There are different approaches that are generally adopted to carry out efficient preprocessing such as Filtering, Discrete Wavelet Transform, and Independent Component Analysis - to name a few [1], [5]–[12]. The feature extraction step transforms the useful information from the signals in the form of the feature vector. There are several ways for extracting features from physiological signals, such as Short Time Fourier Transform (STFT), i.e.,

dividing signals into equal-length segments and extracting statistical features like Power Spectral Density (PSD) and Spectral Entropy (SE), which are widely used for emotion recognition. Another way may be using empirical mode decomposition (EMD), which has the benefit of decomposing non-stationary and non-linear signals while keeping the signal-to-noise ratio high. Then, different statistical features can be extracted from the equal-length segments of the signals. Autoencoder can be used to extract features from non-stationary physiological signals. Its architecture consists of an encoder network and a decoder network, and its training is done to minimize the error between the input data and the reconstructed data.

The feature extraction phase may end in some irrelevant features, so it is necessary to select discriminant features for the recognition problem to improve the recognition model's performance by increasing generalization error and decreasing over-fitting. Moreover, selecting relevant features will ease the process of feature analysis and model training. Different techniques can be used to select features, such as the ReliefF algorithm, Sequential Backward Selection (SBS), and Principal Component Analysis (PCA). Some of the algorithms (ReliefF, SBS) reduce dimensionality by removing some redundant and irrelevant features, and some algorithms (PCA, ICA) transform the original features into new forms of features [13]. Feature fusion can be done in two different ways, one is feature level fusion and other is decision level fusion. In feature-level fusion, features extracted from different signals are combined to get the combined feature set before giving it to the classifier. There are different machine learning-based classifiers that have been used for this purpose, such as Linear Discriminant Analysis (LDA), Quadratic Discriminant Analysis (QDA),  $k$ -nearest neighbor (KNN), and Support Vector Machine (SVM) [6].

Different physiological signals like EEG, ECG, or EMG can be represented as time series data. Each sample in the dataset corresponds to a measurement made at a particular point in time. Preprocessing these signals can be used to normalize the data, filter artifacts, and eliminate noise. Then, features, which can be spatial or temporal domains, are extracted from the signals. Features are then given as input to Deep Neural Networks (DNN) [14], [15]. Each of the one or more hidden layers that make up the DNN has many neurons. Due to these hidden layers, the network may learn sophisticated mappings between the input data and target emotions. ReLU (Rectified Linear Unit) and sigmoid activation functions add non-linearity to the network, enabling it to simulate more complex interactions. The output layer of the DNN typically consists of one or more neurons, each corresponding to a specific emotion class. For binary classification, a single neuron with sigmoid function is used, however, for multi-class classification softmax activation function is used for the classification of emotional classes. Other deep learning architectures known as transformers have become more popular, particularly for Natural Language Processing (NLP) applications. By analyzing textual data (such as social media

postings, reviews, or transcripts) that contains emotional expressions, transformers can be used in emotion identification tasks [16].

Multimodal signals contain a lot of useful information regarding human intuitions. Moreover, different physiological responses are made even within the same emotional state. Therefore, there is a need to address inter-subject variability through emotion recognition models. In addition, certain emotions have the same physiological responses, which creates lots of ambiguity in recognizing the distinct emotions. In the same context, extracting appropriate and discriminant features from physiological signals is a challenging task as well. It is immensely necessary to use the most discriminant features from the physiological signals, which can help analyze the anatomy of different physiological reactions to different stimuli and hence recognize the accurate emotional state. In literature, different hand-crafted and automated features [17]–[24] have been extracted to study the physiological response to different emotional states, but still, there is room to improve the understanding of sophisticated underlying physiological processes related to different human emotions. The main contributions of this research are summarized as follows:

- Effective preprocessing of physiological signals, i.e., EEG, ECG, and GSR, which is robust against complex noise. This was achieved through independent component analysis followed by DWT.
- Developed an effective preprocessing technique for physiological signals (EEG, ECG, and GSR) that is robust against complex noise, achieved through independent component analysis followed by discrete wavelet transform (DWT).
- Extracted a comprehensive feature set by integrating features from multiple deep learning-based techniques.
- Proposed customized architectures for feature extraction using stacked autoencoder, convolutional neural network (CNN), and Vision Transformer.
- Introduced an ensemble learning approach for classifying ECG, EEG, and GSR signals to perform emotion recognition, utilizing both majority voting and Model Agnostic Meta Learning (MAML) to combine the outputs of various machine learning and deep learning classifiers.
- Proposed method achieves high accuracy, sensitivity, and specificity in emotion recognition using physiological signals.

Rest of the paper is organized as follows: Section II presents the critical analysis of the existing state-of-the-art methods of emotion recognition using physiological signals, the proposed method is described in section III, results are presented in the section IV, whereas section V concludes this research and provides future directions.

## II. RELATED WORK

Emotion recognition is one of the hot topics in recent research and has been widely studied under human-computer

interface and affective computing [2], [25]–[27]. Affective computations mainly address pattern recognition for the analysis of emotions [28]. In literature, a lot of effort has been made to recognize human emotions intelligently. Some of them are based on non-physiological signals, which include face recognition, speech recognition, and posture recognition, but these methods are not the actual interpretation of human mental state because they can be manipulated [29]–[31]. However, on the other hand, physiological signals such as respiration rate, Electrocardiogram (ECG), Electroencephalogram (EEG), Galvanic Skin Response (GSR), and Blood Volume Pulse (BVP) have been used for emotion recognition, which are proven to be the effective mean for emotion recognition because they give reliable results as they can't be forged.

Physiological processes such as eye movements, respiration, heartbeat, and head movements cause physiological artifacts. Some common types of noise include Powerline Interferences, Muscle Artifacts, Unintentional muscular contractions or movements during signal recording, and Motion Artifacts. AMIGOS, DEAP, SEED, and DREAMERS datasets were used by Topic et al. [25], where he created feature maps (topography and holography) from the EEG signals of aforementioned datasets and resulted in noisy images due to interferences present in 14 channels of signals. Due to the varying length of EEG signals, Garg et al. [2] decomposed signals into equal-length samples using an approach of Overlapping Sliding Windows (OSW) where he used 512-sized windows with a shift of 32.

The Singular Spectrum Analysis technique proposed by Sharma et al. [28] is a technique used to decompose EEG and ECG signals into reconstructed components (RCs) using a sliding window approach. Preprocessing by investigating ECG and GSR signal peaks was performed by Granados et al. [17]. EEG signals were down-sampled at 128Hz frequency and were filtered using a high pass filter at 2Hz frequency in the proposed work [32]. GSR and ECG physiological signals from AMIGOS and self-generated datasets were employed in [33] for the evaluation of the proposed methodology. Tung et al. [34] performed multimodal-based emotion recognition using physiological signals. Recognition accuracy was increased effectively via baseline removal or recording of signals without a stimulus in the preprocessing phase of work [35].

A strategy for self-supervised emotion identification based on ECG was suggested by Sarkar et al. [36]. For the purpose of identifying emotions, Dar et al. [37] used physiological data such as Electroencephalogram (EEG), Electrocardiogram (ECG), and Galvanic Skin Response (GSR) signals. Prior to being used in neural networks, these signals needed to be processed. A deep learning-based emotion recognition model using EEG signals was proposed by Zhao et al. [38]. In this work, the preprocessing was carried out using a Z-score transformation. Chen et al. [18] used the AMIGOS dataset for the testing of the proposed emotion recognition model. AMIGOS dataset has GSR, ECG, and EEG signals,

and in this work, separate preprocessing is performed for all physiological signals.

Arjun et al. [39] proposed an emotion recognition method using EEG signals. Dang et al. [40] uses the Moving Target Indication (MTI) method to eliminate background noise and the Variational Modal Decomposition (VMD) method to extract and separate heartbeat and respiration signals from the collected physiological data. Feature extraction techniques can be categorized into traditional machine learning-based methods and deep learning techniques. Deep learning-based approaches, such as convolutional neural networks (CNNs) and recurrent neural networks (RNNs), can automatically learn discriminant features from physiological signals without the effort of manual feature extraction. Garg et al. [2] used Fourier and Wavelet transforms for the extraction of two features, i.e., Normalized Band Power (NBP) and Normalized Wavelet Energy (NWE) from the decomposed signals of EEG, respectively. Entropy-based pattern learning for emotion recognition was performed by Sharma et al. [28]. Entropy-based features such as Information Potential (IP) and Centered Correntropy (CEC) can be computed from EEG/ECG signals to extract discriminating information and assist in emotion recognition. An approach to performing analysis of hand-crafted features using machine learning techniques with automated features was presented by Granados et al. [17]. Tung et al. [34] performed a statistical analysis of physiological signals to bridge the association between human emotions and physiological signals.

Shukla et al. [41], in addition to the time and frequency domain features, time-frequency domain features are also extracted for the EDA signals. Different AMIGOS features from 14 channels of EEG signals were extracted by Tejada et al. [42]. The feature selection techniques were used to assess how many features contribute more towards emotion recognition by observing their recognition rates. A novel convolutional layer called the Scaling layer was introduced in [43], which extracted spectrogram-like features from raw EEG signals. This multi-kernel layer takes a 1D input signal and gives 2D spectrogram-like feature maps of a signal. A self-supervised ECG learning system for identifying emotions was demonstrated by Sarkar et al. [36].

Researchers [44] proposed adaptive low-pass and elliptic band-pass filters, enhancing ECG signal processing and classification accuracy up to 89.6% for certain emotional states. Shin et al. [45] integrated multiple physiological signals, including EEG and ECG, to improve interactive content. Oliveira et al. introduced metaFERA, a meta-framework that supports the rapid prototyping of ERS systems by offering pre-configured building blocks, although its effectiveness is limited by the specificity of physiological signals and stages of the emotional recognition process it supports [46]. Limitations in these methods include their specificity to certain signal types, stages of emotional recognition, difficulty in online acquisition of physiological signals, and increased computational complexity [44]. Proposed data augmentation techniques, while improving classification accuracy, face

challenges such as repeatability issues, low variance, and high bias in augmentation sets [44], [46].

Different time domain and frequency domain features were computed from the preprocessed GSR and ECG signals of the AMIGOS dataset by Chen et al. [18]. A technique for identifying emotions that took advantage of the distinctive properties of EEG data by producing feature maps dubbed TOPO-FM and HOLO-FM was presented by Topic et al. [25]. Granados et al. [17], along with the hand-crafted features, performed automated feature extraction using the DCNN model, which involves consecutive layers of CNN layers and pooling layers to extract features from physiological signals automatically. Scalograms and spectrograms were generated from the recorded signals in the method proposed in [33], as CNN takes 2D data for training. Scalogram represents wavelet transform visually in time axes with scale and coefficient values.

A 3D convolutional neural network was used to extract automated features from the preprocessed signals in the work proposed by Zhao et al. [38]. An EEG-based approach for convolutional neural network-based emotion identification was suggested by Fan et al. [47]. Arjun et al. [39] presented the attention mechanism for EEG in order to perform emotion recognition. In emotion recognition, labeling the given signal to one of the classes is the major task [48]–[50].

Features extracted from CNN were fed to the SVM classifier in the work proposed in [25], where RBF (Radial Basis Function) was used as a kernel in SVM. CNN takes the input features of NBP and NWE to classify emotions into different categories in the work proposed in [2]. After the comprehensive analysis of previous methods proposed for emotion recognition using physiological signals, the following research gaps have been figured out, which need to be addressed to contribute to existing knowledge.

- While facial expressions and gestures are commonly used for emotion recognition, physiological signals like ECG and EEG provide a more reliable measure as they are less susceptible to manipulation.
- The significant roles of ECG and GSR in emotion recognition are less presented in literature, despite their strong correlations with emotional states.
- Current research on emotion recognition from physiological signals lacks a deep understanding of how emotions manifest in these signals, calling for further exploration with advanced machine learning techniques.
- Most emotion recognition studies suffer from small sample sizes, limiting the effectiveness and generalizability of the developed models.
- Exploring the interaction between ECG and EEG could reveal crucial insights into how heart and brain rhythms influence emotional regulation [51], [52].
- Emotion recognition studies often focus only on heart rate variability and the QRS complex, overlooking other potentially informative ECG patterns.



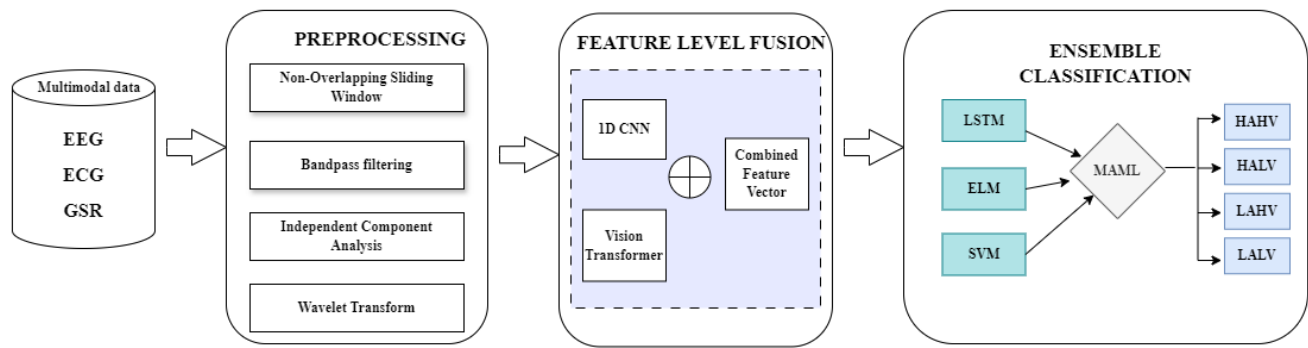


FIGURE 4: Flow diagram of Proposed method for Emotion Charting

### III. MATERIALS AND METHODS

The proposed method presents another deep learning-based approach for emotion recognition, which mainly involves the fusion of discriminant features extracted from the 1D CNN model and Vision Transformer (ViT) along with ensemble classification using MAML (Model Agnostic Meta Learning). The block diagram for this approach is shown in the Figure 4. This approach has three main components, i.e., Preprocessing, Features Extraction, and Classification. Preprocessing involves the segmentation of signals into small segments using a non-overlapping window technique; band-pass filtering is applied to address the removal of noise from the signals. Discrete Wavelet Transform is applied on the output of Independent Component Analysis and then fed into the Vision transformer to extract features, whereas the output of ICA is directly fed into 1DCNN to extract features. Both features extracted from Vision Transformer are merged to make a single feature vector for classification. In classification, an ensemble classifier is used, which takes the probabilities generated by three classifiers, i.e., Long Short-Term Memory (LSTM), Extreme Learning Machine (ELM), and Support Vector Machine (SVM), and generates a final output of emotions classification using a MAML (Model Agnostic Meta Learning) technique.

In this approach, the first physiological signals are passed through non-overlapping sliding windows in order to decompose them into small segments. Then, Bandpass filtering is applied to the segments of signals, and then processed signals are given to the Independent Component Analysis (ICA) module. ICA tends to remove the complex noise from physiological signals such as baseline and powerline noises [2]. After that, Wavelet transform is applied to the cleaned signals to convert them into the frequency domain for further use in feature extraction. The overview of preprocessing is illustrated in Figure 5.

In preprocessing, physiological signals are first decomposed into 30-second-sized windows using non-overlapping sliding windows. The portion of the signal that the window did not cover was trimmed or left out of the calculation. The sizes of the window and shift were computed empirically.

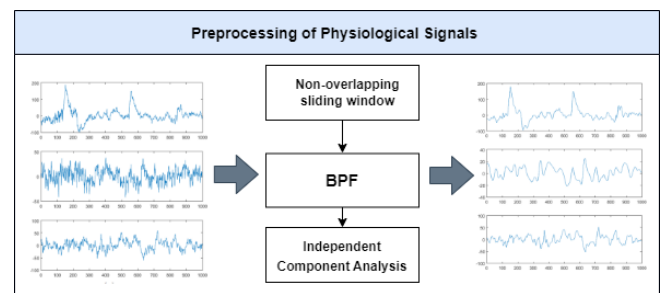


FIGURE 5: Flow diagram of Preprocessing of CVT-based Emotion Charting

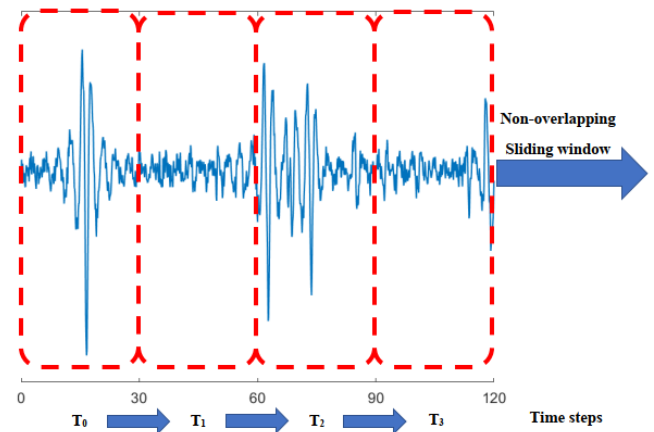
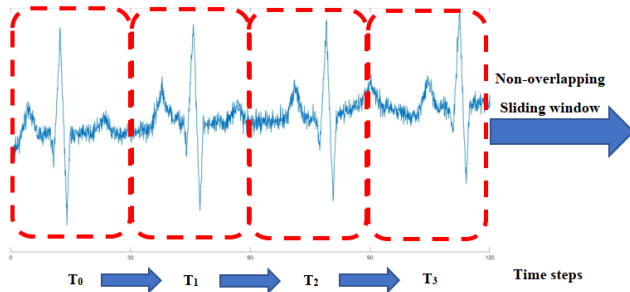


FIGURE 6: Non-overlapping sliding window applied on EEG signal

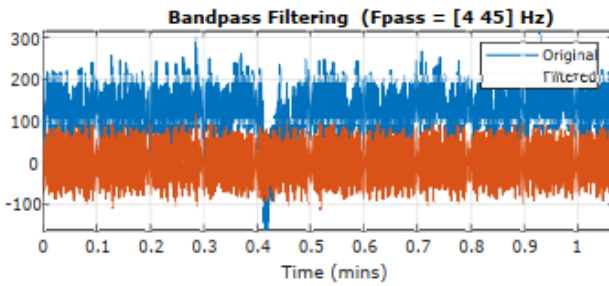
Figure 6 and Figure 7 illustrate EEG and ECG signals after applying a non-overlapping sliding window, respectively.

Due to the baseline shift that occurs in physiological signals during the acquisition of these signals, complex noise is added to them. Therefore, bandpass filters are applied with frequency ranges of 0.5Hz to 45Hz on ECG and EEG signals for the removal of powerlines and baseline noise. Similarly, a bandpass filter with a cutoff frequency of 0.04Hz to 0.25Hz is applied to GSR signals for artifact removal. The output signals of bandpass filtering from EEG in the time domain

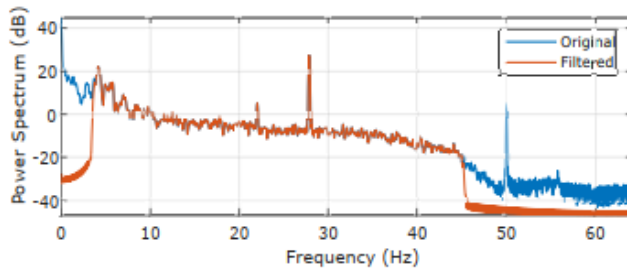


**FIGURE 7:** Non-overlapping sliding window applied on ECG signal

are shown in Figure 8 and the output of bandpass filtering in the frequency domain is shown in Figure 9. Figure 8 shows signals at high frequency have been eliminated after applying the bandpass filter, and Figure 9 shows the frequency spectrum of signals where frequency ranges 0.5Hz to 45Hz will pass, and the noisy components of high frequencies have been filtered after applying the bandpass filter with a cutoff of 0.5Hz to 45Hz.



**FIGURE 8:** Bandpass Filtering of EEG signals in the Time Domain



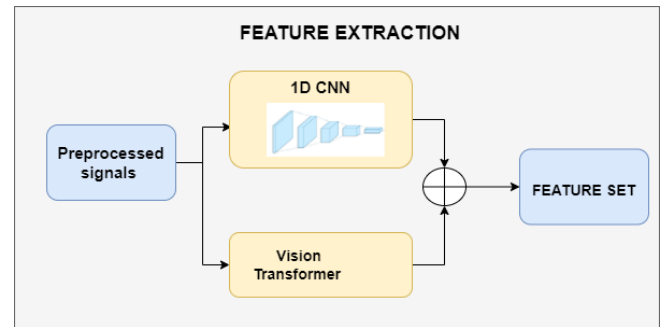
**FIGURE 9:** Bandpass Filtering of EEG signals in the Frequency Domain

Then filtered physiological signals are sent to Independent Component Analysis (ICA), which transforms the signal into the signal having mutually independent components [53]. Hence, the independent components cannot infer information from each other. Independent component analysis is performed on signals in a way similar to the first approach.

After performing the denoising of physiological signals using Independent Component Analysis (ICA). Afterward,

DWT is applied to the ICA output as done in the previous approach to get the Scalograms of the signals. The output of ICA is given to the 1D-CNN feature extractor, and the Scalograms are given to the Vision Transformer-based feature extractor. The mechanism of feature extraction using 1D-CNN and Vision Transformer is discussed in the next section.

After performing preprocessing, the output of ICA is given to the 1D CNN model to extract automated features. As the physiological signals are continuous and have long-term repetitive patterns, a 1D convolution feature detector can make use of parameters sharing, i.e., features learned at one part of the signal can be used in other parts of the signal as well, and features learned for one channel can be used for other channels as well. However, the Scalograms obtained from DWT in preprocessing are given to Vision Transformer to get another feature vector. Because a Vision Transformer makes use of the model's previously learned information gleaned from a sizable dataset, using it as a feature extractor can be advantageous. When you have a limited amount of labeled data for your particular job, this frequently yields potent and semantically significant picture characteristics that can improve the performance of many computer vision tasks. The feature vector obtained from Vision Transformer is then combined with a feature vector obtained from 1D CNN to get a single feature vector, which is then fed to the ensemble classifier for emotion classification. Figure 10 shows the overview of Feature Extraction using 1D-CNN and Vision Transformer.



**FIGURE 10:** Feature Extraction using 1D CNN and Vision Transformer

Figure 11 shows the architecture of the proposed 1D CNN. The input to the 1D-CNN is  $14 \times (128 \times 30)$ , and there are a total of 14 channels of EEG, and each channel contains 128 values of 1 sec. The hyperparameters used for the proposed architecture of 1D-CNN are shown in Table 1, which specify empirically computed values of different hyperparameters for the customized architecture of 1D CNN. However, Table 2 represents the details of activations, weights, and learnable parameters.

The proposed architecture for Vision Transformer is shown in Figure 12. In our proposed approach, Vision Transformer takes Scalograms as input, which are generated by DWT in a preprocessing step. The ViT works on the basis of the attention mechanism by correlating the nearby and long-

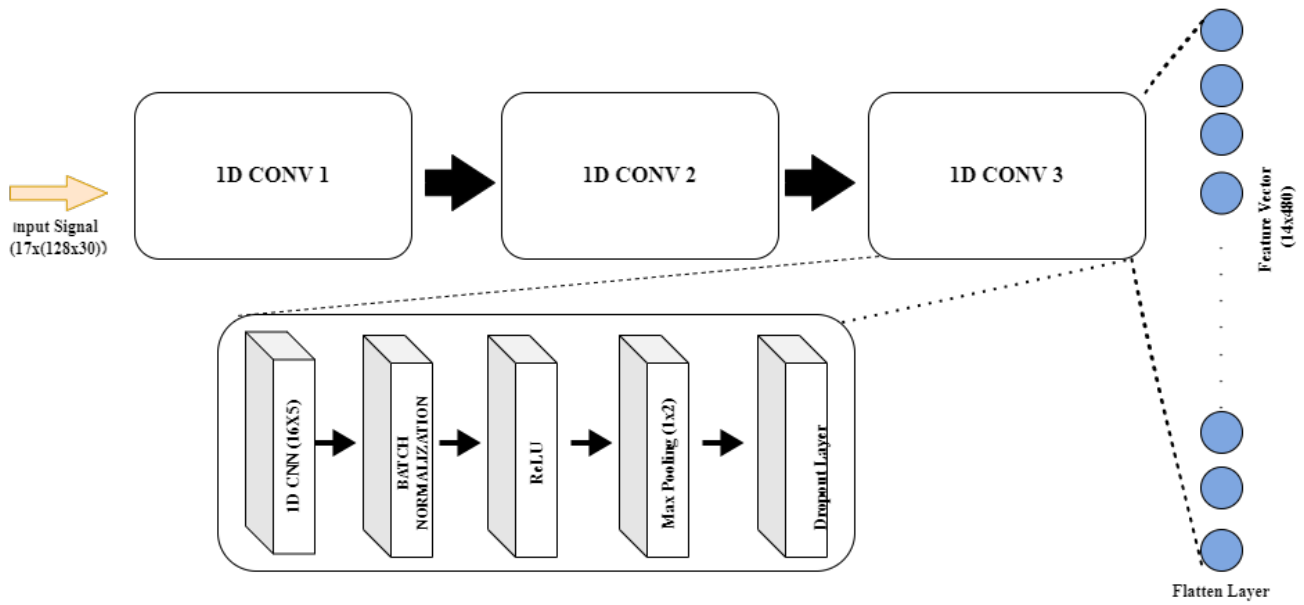


FIGURE 11: Proposed architecture of 1D CNN

TABLE 1: Hyperparameters for the proposed architecture of 1D CNN

| Hyperparameter         | Value                    |
|------------------------|--------------------------|
| Activation Function    | ReLU                     |
| Loss Function          | Categorical_crossentropy |
| Optimizer              | Adam                     |
| Learning Rate          | 0.01                     |
| No. of training Epochs | 50                       |
| Batch Size             | 32                       |
| Dropout                | 0.2                      |

distant pixels. The input image is initially divided into non-overlapping patches that are used as tokens for the attention function. This process produces a 4D matrix with the remaining three dimensions, row, column, and depth, which are indexed by batch. This is comparable to a convolution layer utilizing a kernel. Therefore, the image

$$I \in \mathbb{R}^{H \times W \times C} \quad (1)$$

is redesigned to

$$PP \in \mathbb{R}^{N \times P^2 \times C} \quad (2)$$

where  $H$  and  $W$  represents the height and width of the image and  $C$  represents the number of channels. However,  $N$  is the number of patches, which can be calculated as equation 3.

$$N = \frac{H \times W}{P^2} \quad (3)$$

where  $P$  is the patch size. The input image size is  $(227 \times 227)$ , and the patch size is  $(16 \times 16)$  as shown in Table3, which represents the optimum hyperparameters of

TABLE 2: Hyperparameters details of proposed architecture of 1D CNN

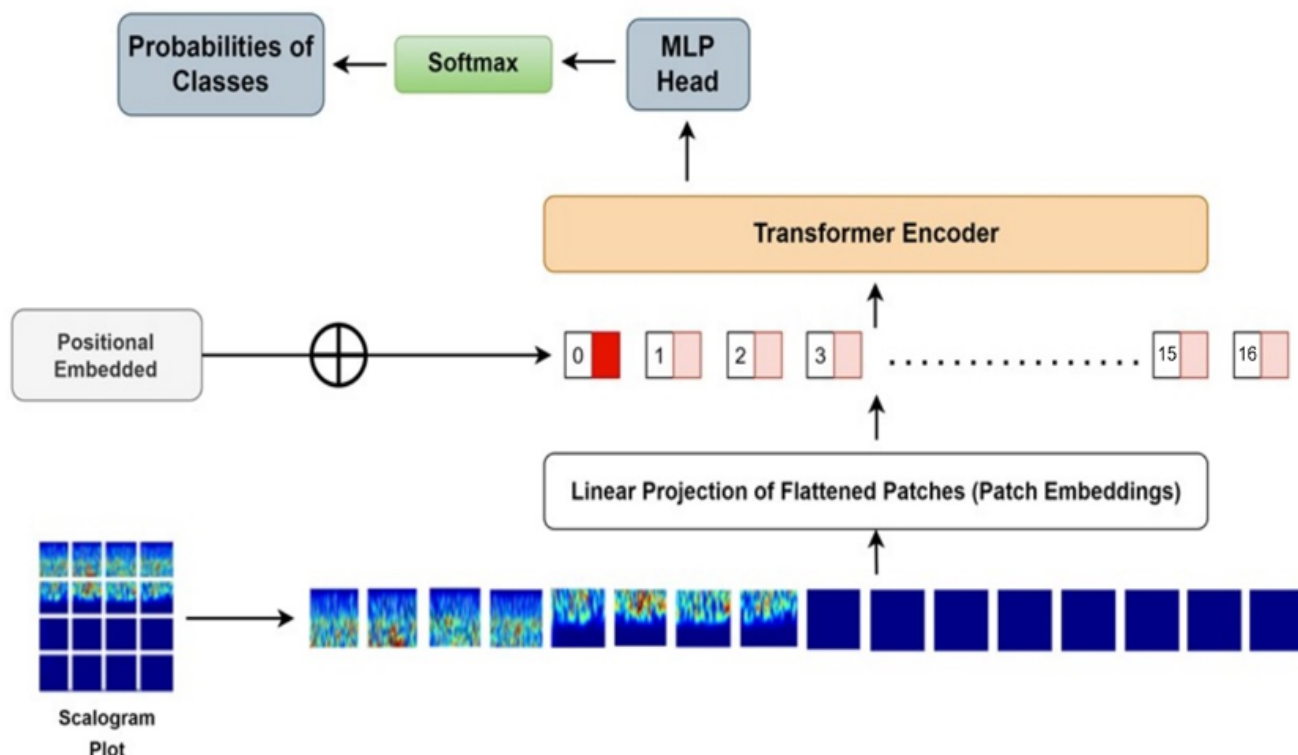
| Name of Layer       | Activations    | Weights  | Parameters |
|---------------------|----------------|----------|------------|
| Input sequence      | 14x128x30      | -        | 0          |
| Conv layer          | 14x(128x30)x16 | 1x8x1x16 | 144        |
| Batch Normalization | 14x(128x30)x16 | 1x1x16   | 32         |
| ReLU                | 14x3840x16     | -        | 0          |
| Max Pooling         | 8x1920x14      | -        | 0          |
| Dropout             | 8x1920x14      | -        | 0          |
| Conv layer          | 24x1920x14     | 1x8x1x16 | 144        |
| Batch Normalization | 24x1920x14     | 1x1x16   | 32         |
| ReLU                | 24x1920x14     | -        | 0          |
| Max Pooling         | 12x960x14      | -        | 0          |
| Dropout             | 12x960x14      | -        | 0          |
| Conv layer          | 28x960x14      | 1x8x1x16 | 144        |
| Batch Normalization | 28x960x14      | 1x1x16   | 32         |
| ReLU                | 28x960x14      | -        | 0          |
| Max Pooling         | 14x480x14      | -        | 0          |
| Dropout             | 14x480x14      | -        | 0          |
| Flatten layer       | 94080          | -        | 0          |

the proposed architecture of Vision Transformer. The number of patches is calculated as:

$$\left( \frac{H \times W}{P^2} \right) = \frac{227 \times 227}{16^2} = 201 \quad (4)$$

After performing a patch partition, the input image is converted into a 2D matrix named  $PP$  and is linearly projected into a 1D embedding vector, which is termed as  $PP_{Linear\_Projection}$ . Table 3 presents the hyperparameters of the proposed architecture of Vision Transformer.

The training approach for single hidden layer feed-forward neural networks (SLFN) called Extreme Learning Machine (ELM) produces promising results with little computational effort as a result of its exceptional performance in numer-



**FIGURE 12:** Block Diagram of Proposed Architecture of Vision Transformer

**TABLE 3:** Hyperparameters for the proposed architecture of Vision Transformer

| Hyperparameter           | Value             |
|--------------------------|-------------------|
| No. of Transformer layer | 12                |
| Hidden dimension layer   | 768               |
| No. of attention heads   | 16                |
| Patch size               | 16x16             |
| Size of input size       | 227x227           |
| Learning rate            | $1 \times e^{-4}$ |
| Batch size               | 32                |
| No. of training epochs   | 50                |
| Weight decay coefficient | 0.05              |

ous real-time learning tasks like classification, regression, clustering, etc. [54]. It is regarded as the most cutting-edge, dependable, and effective method for resolving complicated deep learning-based challenges. For the classification of emotions, it uses the combined feature vector obtained in the preceding phase as an input. Nine thousand hidden neurons are used to train the ELM classifier. For training and validation, the training samples are split 80:20. As an illustration, training samples of the feature vector are further divided into 20% of data for ELM classifier validation and 80% of data (training data) for training the ELM classifier. Validation data is used to validate the ELM classifier after training. Following validation, the performance of the suggested system is evaluated using testing samples from the

feature vector. As previously mentioned, the feature vector obtained in the previous phase is fed to LSTM, ELM, and SVM classifiers. The output of each classifier is given to an ensemble classifier, i.e., MAML (Model Agnostic Meta Learning), which generates classification voting using the majority voting scheme. These classification probabilities are then fed to LSTM-based Model Agnostic Meta Learning for final classification.

In the context of emotion recognition, this approach can be used to train a model using a small amount of data from each emotional state and then quickly adapt the model to classify new examples. The MAML algorithm is a specific implementation of agnostic meta learning that has been used in this system. It combines the probabilities from three different classifiers, which are each trained using a different set of features extracted from the physiological signals. The LSTM-based model is used to aggregate the probabilities from the three classifiers and make a final decision on the emotional state. Ensemble classification approaches like MAML can improve the robustness and accuracy of emotion recognition systems. By combining the outputs of multiple classifiers trained using different feature sets, the system can better capture the complex patterns within the physiological signals related to changes in emotional state.

To implement this approach, a multilayer perceptron with two hidden layers is used as the meta-learner. This MLP has 256 and 64 neurons in the first and second hidden layers,



**TABLE 4:** Description of AMIGOS and DEAP datasets

| Dataset | Participants | Trials/<br>Video clips | Channels | Experiments | Affective states | threshold | Sampling<br>frequency<br>(Hz) |
|---------|--------------|------------------------|----------|-------------|------------------|-----------|-------------------------------|
| DEAP    | 32           | 40                     | 32       | 1           | Valence/Arousal  | 1-9 (4.5) | 128                           |
| AMIGOS  | 40           | 16                     | 14       | 2           | Valence/Arousal  | 1-9 (4.5) | 128                           |

respectively, and 2 output neurons for classification. Three classifiers, SVM, ELM, and LSTM, are used to generate probabilities for each emotional category. These probabilities are then fed into the MAML model for classification. The output of the MAML model identifies four different categories of emotions, which are HALA, LAHV, LALV, and HAHV. By combining the outputs of multiple classifiers using MAML, the proposed system is able to achieve high accuracy in emotion recognition using physiological signals.

#### IV. RESULTS AND DISCUSSION

We evaluate the effectiveness of the proposed methods of emotion charting by carrying out comprehensive experiments. The validation of this work is assessed by performing an ablation study with a number of experimentations for three main phases of the proposed system: preprocessing, Feature Extraction, and Classification. At the end of each ablation study, the best techniques for the particular phase were chosen based on the proposed approaches by analyzing the results.

##### A. DATASETS

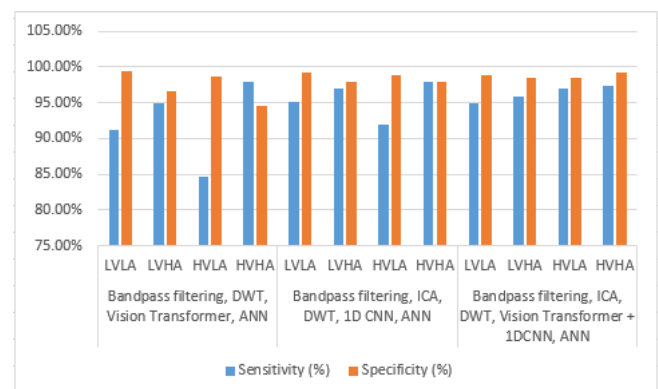
This research is validated on two datasets: AMIGOS (a dataset for Affect, Personality, and Mood research on Individuals and Groups) and DEAP (a Dataset for Emotion Analysis using Physiological signals). A multimodal dataset called DEAP (Dataset for Emotion Analysis using Physiological signals) combines blood volume pulse, electrooculogram (EOG), electromyography (EMG), and EEG. 32 persons, 16 men and 16 women, ages 19 to 37 (mean age 26.9), with skin temperature, plethysmograph readings, and galvanic skin responses, were utilized to gather the data. A general comparison of both datasets is given in Table 4.

##### B. EXPERIMENTAL RESULTS

An ablation study was conducted to devise a system for efficient feature extraction and classification for Emotion Charting. Preprocessing techniques have been kept the same that includes bandpass filtering and DWT. Therefore, a series of experiments have been conducted to carry out the ablation study for feature extraction and classification. In the feature extraction phase, Method 1 refers to the feature extraction using 1D CNN, Method 2 refers to the feature extraction using Vision Transformer, and Method 3 involves the fusion of feature vectors obtained from 1D CNN and Vision Transformer. However, in the classification phase, the best feature vector obtained from the previous step is fed

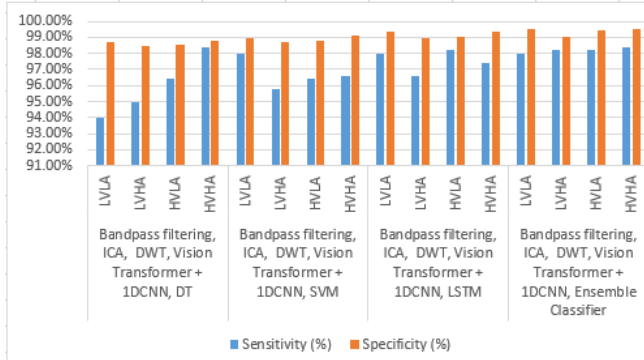
**TABLE 5:** Experimental Results of the ablation study using CVT approach for Emotion Charting (with AMIGOS dataset)

| Methodology                                                                   | Class Label | Acc (%) | Sen (%) | Spec (%) |
|-------------------------------------------------------------------------------|-------------|---------|---------|----------|
| Bandpass filtering, DWT, Vision Transformer, ANN                              | LVLA        | 92.2    | 91.20   | 99.33    |
|                                                                               | LVHA        |         | 95.00   | 96.54    |
|                                                                               | HVLA        |         | 84.60   | 98.61    |
|                                                                               | HVHA        |         | 98.00   | 94.62    |
| Bandpass filtering, ICA, DWT, 1D CNN, ANN                                     | LVLA        | 95.55   | 95.20   | 99.20    |
|                                                                               | LVHA        |         | 97.00   | 98.01    |
|                                                                               | HVLA        |         | 92.00   | 98.77    |
|                                                                               | HVHA        |         | 98.00   | 97.93    |
| Bandpass filtering, ICA, DWT, Vision Transformer + 1DCNN, ANN                 | LVLA        | 96.30   | 95.00   | 98.80    |
|                                                                               | LVHA        |         | 95.80   | 98.44    |
|                                                                               | HVLA        |         | 97.00   | 98.56    |
|                                                                               | HVHA        |         | 97.40   | 99.17    |
| Bandpass filtering, ICA, DWT, Vision Transformer + 1DCNN, DT                  | LVLA        | 95.95   | 94.00   | 98.73    |
|                                                                               | LVHA        |         | 95.00   | 98.43    |
|                                                                               | HVLA        |         | 96.40   | 98.56    |
|                                                                               | HVHA        |         | 98.40   | 98.75    |
| Bandpass filtering, ICA, DWT, Vision Transformer + 1DCNN, SVM                 | LVLA        | 96.70   | 98.00   | 98.93    |
|                                                                               | LVHA        |         | 95.80   | 98.71    |
|                                                                               | HVLA        |         | 96.40   | 98.78    |
|                                                                               | HVHA        |         | 96.60   | 99.11    |
| Bandpass filtering, ICA, DWT, Vision Transformer + 1DCNN, LSTM                | LVLA        | 97.55   | 98.00   | 99.33    |
|                                                                               | LVHA        |         | 96.60   | 98.99    |
|                                                                               | HVLA        |         | 98.20   | 99.05    |
|                                                                               | HVHA        |         | 97.40   | 99.32    |
| Bandpass filtering, ICA, DWT, Vision Transformer + 1DCNN, Ensemble Classifier | LVLA        | 98.20   | 98.00   | 99.53    |
|                                                                               | LVHA        |         | 98.20   | 99.06    |
|                                                                               | HVLA        |         | 98.20   | 99.46    |
|                                                                               | HVHA        |         | 98.40   | 99.53    |

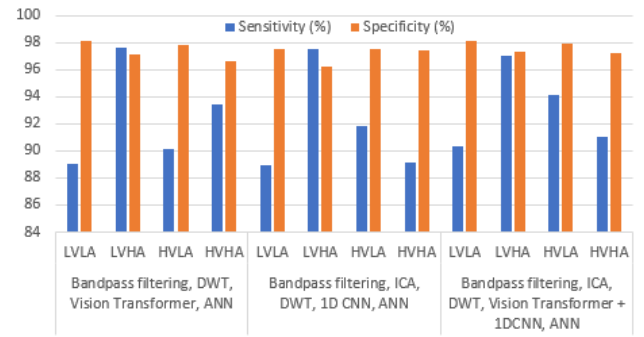


**FIGURE 13:** Performance Evaluation of CVT-based Emotion Charting with AMIGOS dataset for Feature Extraction

to the ensemble classifier where Method 1 corresponds to classification using Decision Tree, Method 2 corresponds to classification using Support Vector Machine, Method 3 corresponds to classification with Long Short Term Memory and Method 4 employs ensemble learning by combining the



**FIGURE 14:** Performance Evaluation of CVT-based Emotion Charting with AMIGOS dataset for Classification

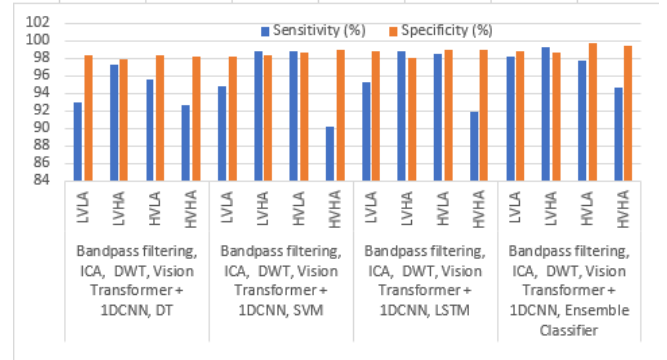


**FIGURE 15:** Ablation Study: Performance Evaluation of CVT-based Emotion Charting with DEAP dataset in terms of Sensitivity for Feature Extraction

effects of all above classifiers.

Table 5 shows the experimental results of the ablation study on the AMIGOS dataset. Figure 17 presents the confusion matrices of all experiments of ablation study on AMIGOS dataset. First of all, the preprocessed signal obtained after performing bandpass filtering and DWT (as done in the previous approach) is fed to the Vision Transformer for feature extraction. In this experiment, the system achieved an accuracy of 92.20% with a maximum sensitivity of 98.0%. In the second experiment, the output of DWT followed by ICA is fed to 1D CNN for feature extraction, and the system achieved a better accuracy of about 95.5%. However, the third experiment combines the feature vector generated from 1D CNN and Vision Transformer, and the system achieved the highest accuracy of 96.30%. Therefore, the features level fusion of the feature vector generated from 1D CNN and Vision Transformer has generated better results, so this feature vector has been used for different classifiers.

First of all, a combined feature vector is fed to the Decision Tree for classification, and the system obtained an accuracy of 95.9%. The system obtained an accuracy of 96.70% and 97.55% with SVM and LSTM, respectively. However, the system obtained the highest accuracy of 98.20% and a sensitivity of 98.40% with the ensemble classifier, i.e., the MAML technique, which combines the output of DT, SVM,



**FIGURE 16:** Ablation Study: Performance Evaluation of CVT-based Emotion Charting with DEAP dataset in terms of Specificity for Classification

**TABLE 6:** Experimental Results of the ablation study using CVT approach for Emotion Charting (with DEAP dataset)

| Methodology                                                                   | Class Label | Acc (%) | Sen (%) | Spec (%) |
|-------------------------------------------------------------------------------|-------------|---------|---------|----------|
| Bandpass filtering, DWT, Vision Transformer, ANN                              | LVLA        | 92.5    | 89.0    | 98.10    |
|                                                                               | LVHA        |         | 97.5    | 97.1     |
|                                                                               | HVLA        |         | 90.14   | 97.8     |
|                                                                               | HVHA        |         | 93.40   | 96.6     |
| Bandpass filtering, ICA, DWT, 1D CNN, ANN                                     | LVLA        | 91.57   | 88.9    | 97.5     |
|                                                                               | LVHA        |         | 97.5    | 96.1     |
|                                                                               | HVLA        |         | 94.1    | 97.8     |
|                                                                               | HVHA        |         | 91.0    | 97.1     |
| Bandpass filtering, ICA, DWT, Vision Transformer + 1DCNN, ANN                 | LVLA        | 93.10   | 90.29   | 98.10    |
|                                                                               | LVHA        |         | 97.00   | 97.28    |
|                                                                               | HVLA        |         | 94.14   | 97.89    |
|                                                                               | HVHA        |         | 91.0    | 97.1     |
| Bandpass filtering, ICA, DWT, Vision Transformer + 1DCNN, DT                  | LVLA        | 94.60   | 93.00   | 98.30    |
|                                                                               | LVHA        |         | 97.29   | 97.81    |
|                                                                               | HVLA        |         | 95.57   | 98.20    |
|                                                                               | HVHA        |         | 92.5    | 98.1     |
| Bandpass filtering, ICA, DWT, Vision Transformer + 1DCNN, SVM                 | LVLA        | 95.51   | 94.7    | 98.1     |
|                                                                               | LVHA        |         | 98.7    | 98.2     |
|                                                                               | HVLA        |         | 98.7    | 98.5     |
|                                                                               | HVHA        |         | 90.14   | 98.91    |
| Bandpass filtering, ICA, DWT, Vision Transformer + 1DCNN, LSTM                | LVLA        | 96.07   | 95.2    | 98.7     |
|                                                                               | LVHA        |         | 98.7    | 98.04    |
|                                                                               | HVLA        |         | 98.4    | 98.9     |
|                                                                               | HVHA        |         | 91.8    | 98.9     |
| Bandpass filtering, ICA, DWT, Vision Transformer + 1DCNN, Ensemble Classifier | LVLA        | 97.4    | 98.1    | 98.8     |
|                                                                               | LVHA        |         | 99.2    | 98.5     |
|                                                                               | HVLA        |         | 97.7    | 99.6     |
|                                                                               | HVHA        |         | 94.5    | 99.4     |

and LSTM for classification. Figure 13 and 14 show the performance of the ablation study of the CVT-based approach (with AMIGOS dataset) in terms of sensitivity and specificity for feature extraction and classification, respectively.

Table 6 shows the experimental results of an ablation study on the DEAP dataset using a based approach for Emotion Charting. First of all, the output of DWT, as performed in the previous approach, is fed to the Vision Transformer for feature extraction, and the system achieved an accuracy of 92.5%. In the second experiment, the output of ICA was fed to 1D CNN for feature extraction, and the system achieved a better accuracy of about 91.57%. However, in the third experiment, the combined feature vector obtained from 1D CNN and Vision Transformer was fed to ANN, and the system achieved the highest accuracy of 93.10%, which is comparatively a better result. So, the combined feature is then fed

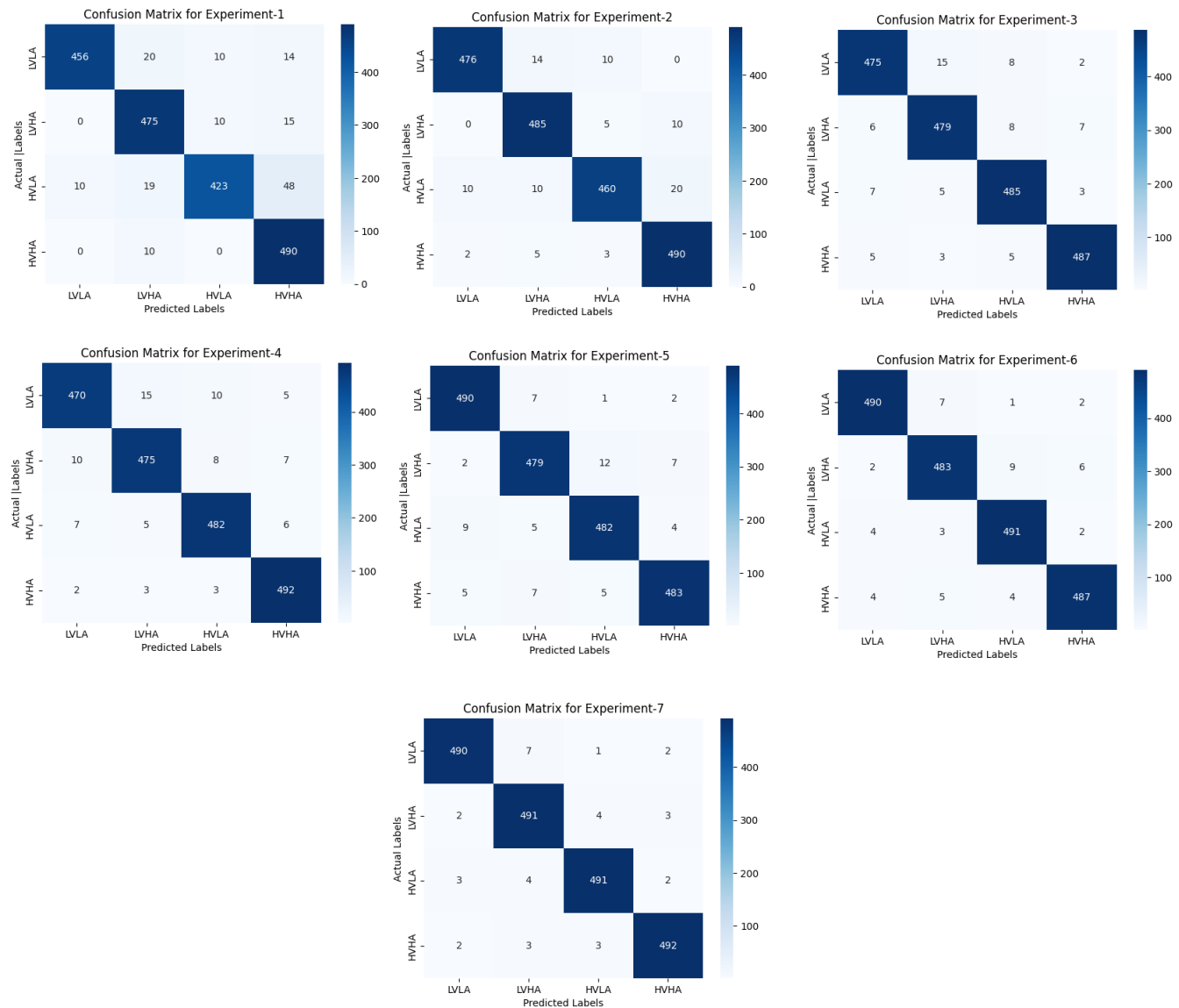


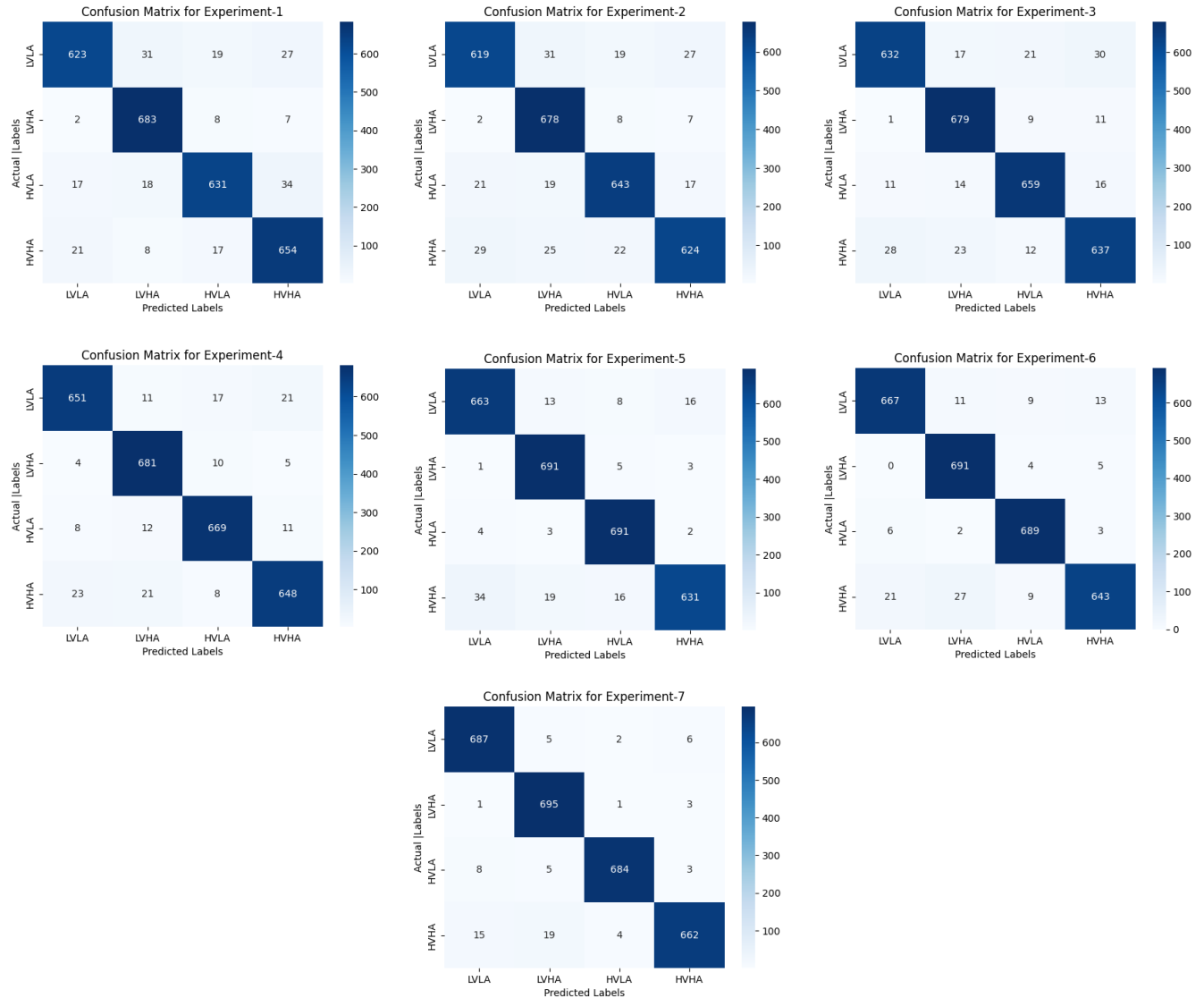
FIGURE 17: Confusion matrix of multiple experiments of ablation study on AMIGOS dataset

to multiple classifiers to carry out the classification ablation study. A combined feature vector is fed to the Decision Tree for classification, and the system obtained an accuracy of 94.6%. Then SVM and LSTM gave the accuracy of 95.5% and 96.01%, respectively. However, the system obtained the highest accuracy of 97.4% with the ensemble classifier, i.e., the MAML technique, which combines the output of DT, SVM, and LSTM for classification. Figure 15 and 16 shows the performance of the ablation study of the CVT based approach (with DEAP dataset) in terms of sensitivity and specificity for feature extraction and classification respectively. Figure 18 presents the confusion matrices of ablation studies performed on DEAP dataset.

It is evident from the literature review that the proposed CVT-based emotion recognition gives the highest accuracy of 98.2% as compared to other state-of-the-art methods of

emotion recognition. The CVT-based approach involves a fusion of features extracted from CNN and Vision Transformer, where CNN involves automated temporal feature extraction, and its customized architecture involves max pooling and dropout layers, which enhance the generalization power of the system. Moreover, Vision Transform involves a self-attention mechanism, which gives the system better performance with combined feature vectors. On the other hand, ensemble learning involves gauging the strength of different sophisticated classifiers. Therefore, these factors contribute towards the better performance of the proposed system in comparison with previously proposed state-of-the-art methods.

It is performing better than other approaches because the customized architecture of CNN involves a max pooling layer and dropout layer, which enhance the generalization



**FIGURE 18:** Confusion matrix of multiple experiments of ablation study on DEAP dataset

power of the system with minimal over-fitting, and CNN involves automated temporal features. However, ViT involves self-attention and extracts hierarchical features. It is clearly observed in the table that methods with preprocessing give improved results as compared to methods without preprocessing. Code of the proposed methodology is available at <https://github.com/Amna707/Emotion-Charting-.git> for reproducibility.

## V. CONCLUSION

Emotion recognition is a key area of affective computing, aimed at interpreting human emotions through physiological signals like EEG, ECG, GSR, temperature, and respiration, which are produced by the Autonomous Nervous System (ANS). Recognizing emotions from these signals is complex due to their noise and non-stationary nature, necessitating advanced preprocessing techniques to enhance accuracy. Tradi-

tional methods involve three main steps: preprocessing, feature extraction, and classification. This research introduces two deep learning-based frameworks for emotion recognition. The first framework employs bandpass filtering and Independent Component Analysis (ICA) for noise reduction, uses a stacked autoencoder for feature extraction, and an ensemble classifier combining SVM, Random Forest, and LSTM for classification, achieving an accuracy of 94.5%. The second framework uses a similar preprocessing approach but includes Discrete Wavelet Transform and feature-level fusion with 1D CNN and Vision Transformer, with an ensemble classifier leading to 98.20% accuracy, 98.40% sensitivity, and 99.53% specificity. These frameworks outperform existing methods, though future work could further explore frequency band analysis and additional deep learning models to address current limitations.

Emotion recognition using physiological signals has va-



riety of applications, however, it also raises several ethical concerns. One primary concern is the privacy of participants whose physiological data is being collected and analyzed. To address these ethical considerations, it is made sure in the publicly available datasets that all the participants provided informed consent, and their physiological data was anonymized to protect their identities. Another concern is the potential misuse of this emotion recognition method in manipulative marketing. It has been taken care that these emotion recognition methods are restricted in the applications of healthcare and mental health assessment to benefit the individuals as well as society.

While the proposed method achieves high accuracy in emotion recognition, there are some limitations that need to be addressed in future. First, the publicly available datasets AMIGOS and DEAP used in this research have limitations in terms of sample size and diversity. The relatively small and similar age group 1samples may not fully represent the broader population, that lead towards limiting the generalizability. Proposed approach utilizes advanced signal processing and machine/ deep learning techniques, which require increased computational resources. The hardware used includes high-performance GPUs to handle the processing tasks involved in training and running the models. This poses challenge for deploying the emotion recognition system in resource-constrained environments.

The future work for this research in affective computing, which currently uses EEG, ECG, and GSR signals from AMIGOS and DEAP datasets for high-accuracy emotion recognition, aims to expand into real-time data collection in a controlled environment. This would involve capturing a broader array of physiological signals from both healthy and affected individuals to enhance the depth and accuracy of emotion analysis. Additionally, the integration of data from heart rate, respiration, and other sources could lead to more comprehensive emotion recognition. In the future, optimized wearable devices for continuously monitoring physiological signals and emotion prediction can also be developed. Further, the research could extend to implementing the system on computing hardware like FPGAs, utilizing a lightweight CNN model optimized for resource efficiency through network compression and exploiting parallel processing to enhance performance on edge devices. This approach not only aims to overcome existing limitations but also seeks to harness this technology responsibly in various applications.

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